

Smart Discounting

When Does It Work and When Does It Hurt?

A Data Analysis Case Study
Sample Superstore Dataset | SQL (Big Query) + Tableau Public

Executive Summary

Discounting is one of retail's most common strategies—yet also one of its most dangerous when applied blindly. This analysis investigates whether discounting always benefits a business or whether it reaches a point where it destroys profitability.

Using SQL in Big Query for data analysis and Tableau Public for visualization, we identify the optimal discount threshold and examine how discounting affects profit across categories, regions, and time periods.

Key Finding: Discounting is only effective up to 20%. Beyond that, profit turns negative.

Dataset & Methodology

Attribute	Details
Source	Sample Superstore Dataset
Platform	Big Query (superstore-499203)
Records	9,994 transactions
Period	2014–2017
Tools	SQL (Big Query), Tableau Public

Data Cleaning: Removed duplicate records • Validated discount ranges • Standardized date formats (YYYY-MM-DD) • Verified no NULL values

Research Objectives

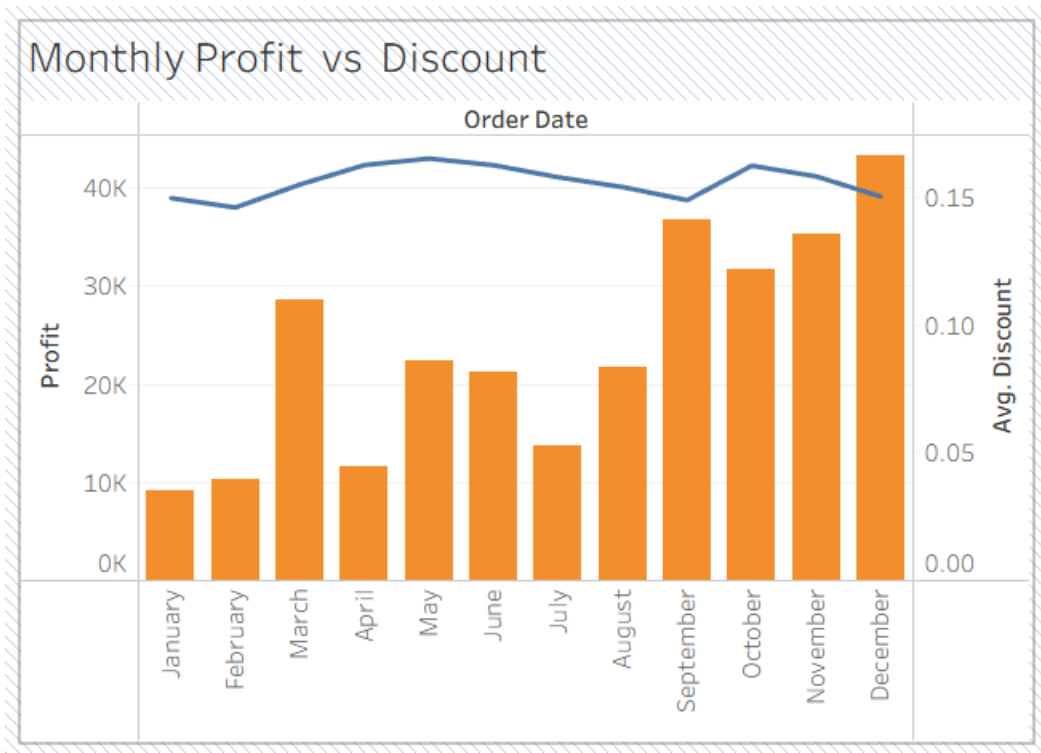
1. **Seasonal Objective:** Identify which months/quarters show that discounting increases profit vs. destroys it.
2. **Category Objective:** Determine which product categories benefit from discounts and which should never be discounted.
3. **Strategy Objective:** Find the optimal discount percentage—the exact point before profit turns negative.

Key Business Questions

- In which month does high discount lead to HIGH profit? (sweet spot months) • In which month does high discount lead to LOSS? (danger zone months)
- Which category should NEVER be discounted?
- Which category BENEFITS most from discounting?
- Is Q4 sales driven by discounts or demand?
- What is the optimal discount % before profit turns negative?
- Which region responds best to discounts?

Analysis & Findings

1. Monthly Profit vs. Discount

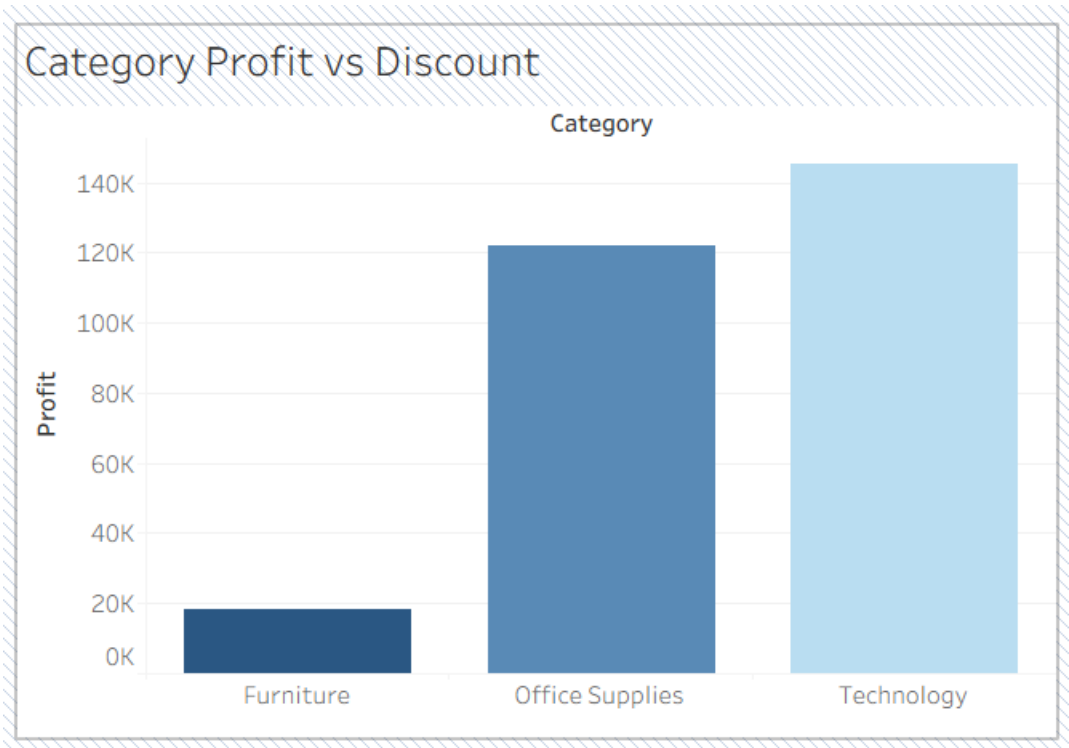


Key Pattern: Average discount stays flat (0.15–0.17), yet profit varies dramatically (\$9k–\$43k).

Metric	Finding
Highest Profit	December (\$43,272), & November (\$35,311)
Lowest Profit	January (\$9,134), February(\$10294), April (\$11,545), July (\$13,791)
Sweet Spot Months	November, December, and March
Danger Zone Months	January, April, and July

■ **Key Insight:** Discounting is not what drives monthly profit. Seasonal demand is the real driver.

2. Category Profit vs. Discount

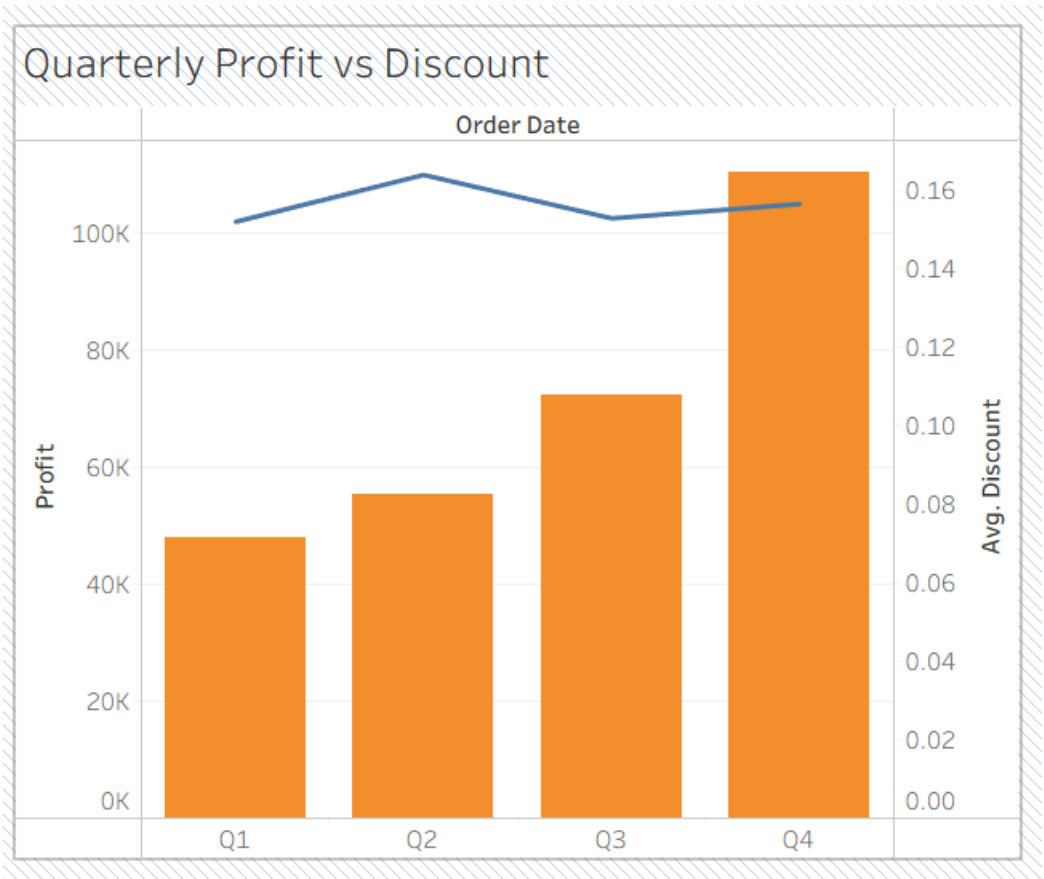


Key Pattern: The more discount given, the less profit earned.

Category	Avg. Discount	Total Profit	Profit Margin	Status
Technology	13%	\$145,386	17.4%	■ Best
Office Supplies	16%	\$122,247	17.02%	✓ Good
Furniture	17%	\$18,380	2.48%	■ Worst

■ **Key Insight:** Furniture is most discounted but least profitable—stop discounting Furniture.

3. Quarterly Profit vs. Discount

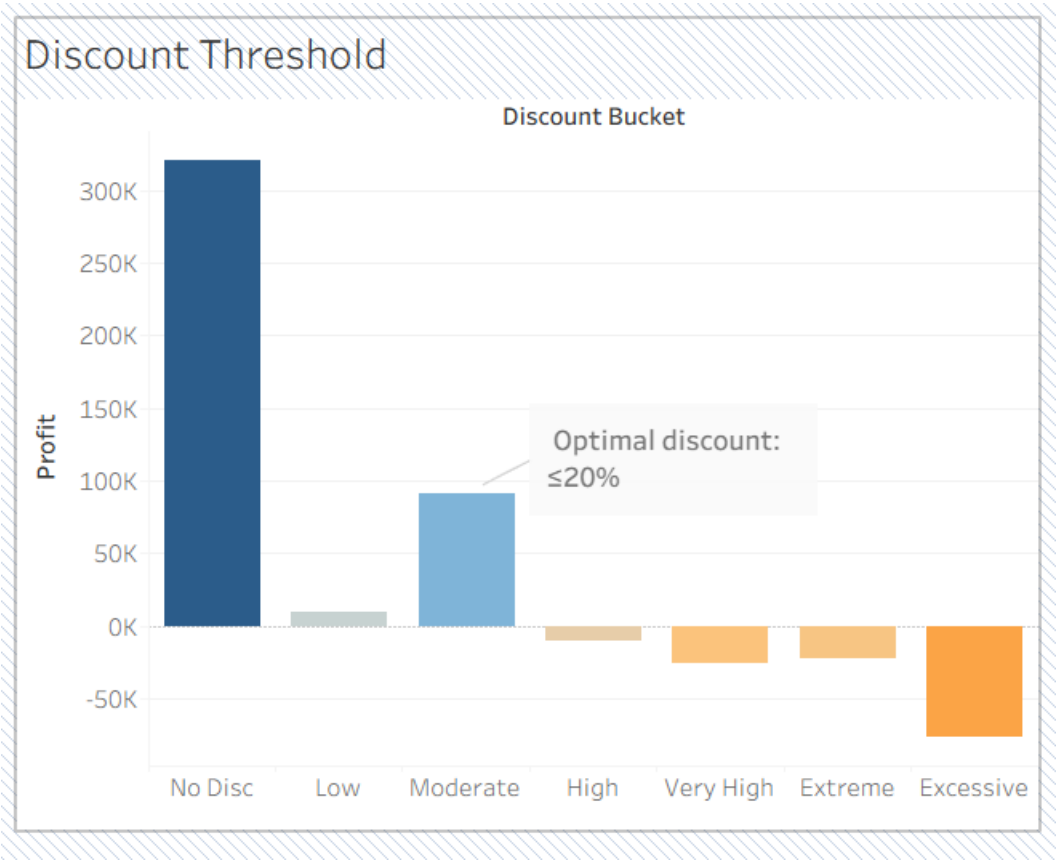


Key Pattern: Q4 generates highest profit, but discounts remain flat across quarters.

Quarter	Avg. Discount	Total Profit	Profit Margin	Transactions
Q4	16%	\$110,367	12.58%	3,695
Q3	15%	\$72,403	11.8%	2,797
Q2	16%	\$55,242	12.42%	2,118
Q1	15%	\$48,003	13.35%	1,376

■ **Key Insight:** Q4 wins on volume, not discounts—promotional spending in Q4 is unnecessary.

4. Discount Threshold Analysis (CRITICAL FINDING)

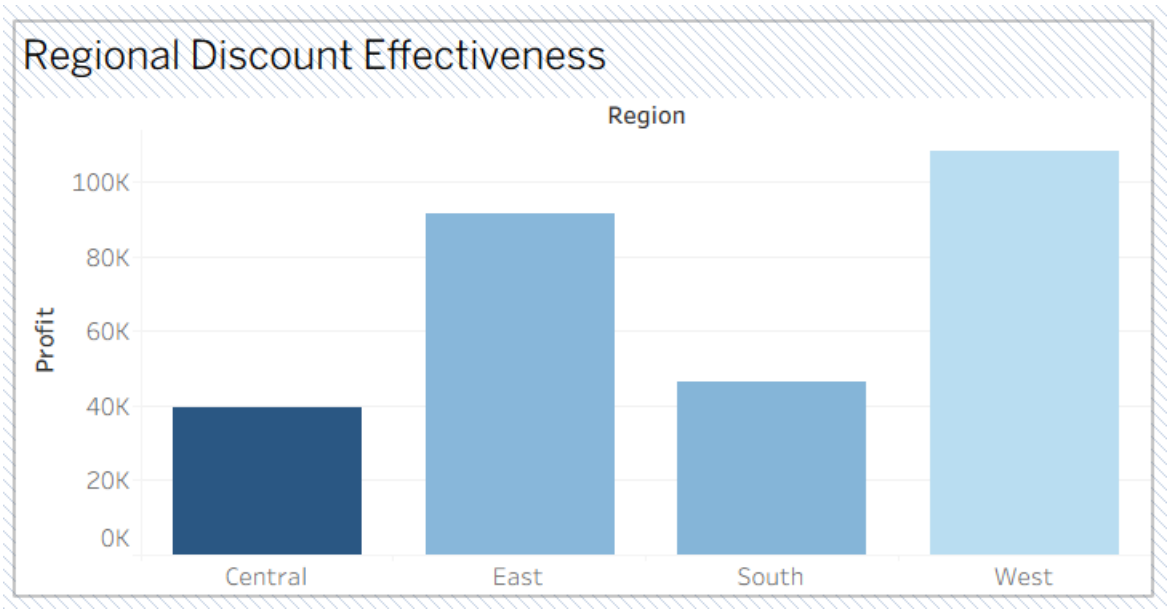


This is the most critical finding. Orders with 0% discount generate \$320,661 profit (29.5% margin). Profit remains positive through 11–20% (\$91,687, 11.6% margin). But beyond 20%, profit turns sharply negative. The 21–30% bucket shows a loss of \$10,357 (-10.1% margin). Losses worsen at every subsequent level.

Discount Range	Avg. Discount	Total Profit	Profit Margin	Status
0% (No Discount)	0%	\$320,661	29.5%	✓ Optimal
1–10%	10%	\$9,029	16.61%	✓ Profitable
11–20%	20%	\$91,687	11.58%	✓ Profitable
21–30%	30%	-\$10,357	-10.1%	■ Loss
31–40%	39%	-\$25,448	-19.44%	■ Loss
41–50%	49%	-\$22,999	-35.71%	■ Loss
50%+	72%	-\$76,559	-119.2%	■ Loss

■ **Key Insight:** 20% is where discounting stops being a strategy and starts being a loss.

5. Regional Discount Effectiveness



Key Pattern: Lower average discounts correlate with higher profits.

Region	Avg. Discount	Total Profit	Profit Margin	Transactions
West	11%	\$108,404	14.94%	3,202
East	15%	\$91,353	13.48%	2,845
South	15%	\$46,549	11.9%	1,616
Central	24%	\$39,706	7.92%	2,323

■ **Key Insight:** *West's low-discount strategy is most effective. Central's heavy discounting requires urgent review.*

Summary of Findings

Business Question	Answer
Sweet spot months?	Nov & Dec (and March)
Danger zone months?	Jan, Apr, Jul
Never discount?	Furniture (17% discount, \$18k profit, 2.48% margin)
Benefits most?	Technology (13% discount, \$145k profit, 17.4% margin)
Q4 driven by?	Demand (not discounts)
Optimal threshold?	20%—profit turns negative beyond
Best region?	West (11% discount, \$108k profit)

Recommendations

1. Cap All Discounts at 20%

Every discount above 20% results in net loss. This is the single most impactful change the business can make.

2. Stop Discounting Furniture

Furniture consistently underperforms with lowest profit and weakest margin. Discounting only destroys margin.

3. Treat Technology and Office Supplies as Primary Promotion Candidates

Both categories maintain strong margins (17%+) and respond well to moderate discounting.

4. Review Central Region's Discount Strategy Immediately

Central gives 24% average discount—highest of any region—yet earns lowest profit (7.92% margin).

5. Do Not Use Discounts to Boost Slow Months

January, April, and July have same discount levels as high-performing months but far lower profit.

6. Use West Region as a Benchmark

West's strategy—low discount, high profit—should be studied and replicated across other regions.

Conclusion

This analysis set out to answer a critical business question: **Does discounting always benefit a business, or can it become a liability?**

Through SQL-based analysis and Tableau visualization, the findings clearly demonstrate that **discounting is only effective up to a certain point—beyond 20%, profit consistently turns negative.**

Key Insights:

- Q4 growth is demand-driven, not discount-driven
- Technology outperforms other categories with minimal discounting
- Furniture suffers from over-discounting with lowest profit returns
- West region achieves highest profitability through moderate, strategic discounting

These insights demonstrate that discounting should be **data-driven, not assumption-driven.** Businesses that monitor the relationship between discount levels and profit are better positioned to sustain long-term profitability.

The data-backed answer to "How much should we discount?" is: 20% or less.

Tools & Skills Demonstrated

SQL (Google Big Query): Data cleaning • Duplicate detection • Window functions • Date parsing • Aggregation • CASE statements • Subqueries

Tableau Public: Dual-axis charts • Calculated fields • Colour encoding • Dashboard design

Business Analysis: Translating raw data into actionable recommendations • Pattern identification • Stakeholder communication